Simple Rules, Complex Behavior

Name _____

Work with a partner. Compare answers when you finish each problem.

Word bank: 2 30 a pattern that repeats chaos emergence the wave triangles

You can use a word more than once.

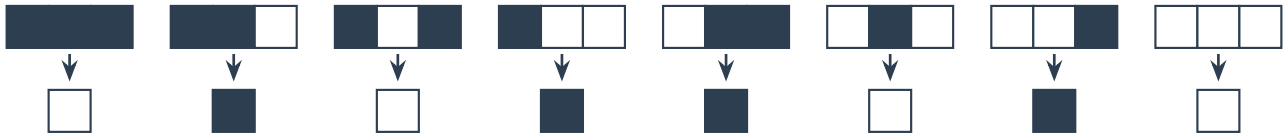
1. The wave. Was anyone in charge of the wave? _____

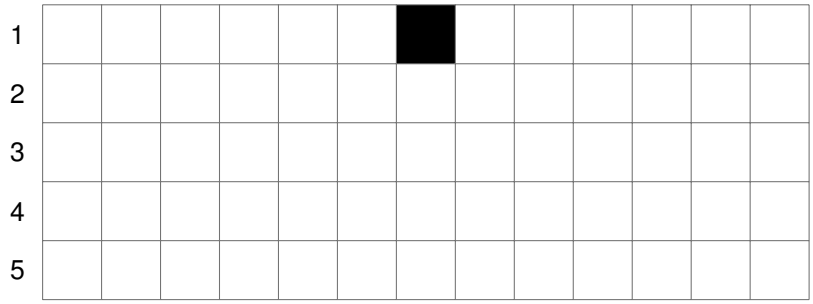
Did the person you watched move down the row? _____ So what moved? _____

When many simple parts make something bigger that nobody planned, it's called _____.

2. Make a few rows. Look at a square and its two neighbors in the row above. Find the same 3 colors on the card. Color the square below. (Outside the grid, everything is white.)

Rule 90: what the middle cell becomes next





What shape is starting to appear? _____

3. One small change. How many answers are different on the Rule 30 card? _____

Rule 90 makes _____. Rule 30 makes _____.

The cone snail's shell looks like Rule _____.

Word bank: 2 or 3 3 backward the pattern

4. Explore the rules. Draw each rule. Circle what it looks like.

Rule	What does it look like?			
110	never changes	a pattern that repeats	chaos	complex
4	never changes	a pattern that repeats	chaos	complex
150	never changes	a pattern that repeats	chaos	complex
30	never changes	a pattern that repeats	chaos	complex

My own rule. My rule number is _____ It looks like _____

Traffic (Rule 184). Every car moves forward. Which way do the jams move? _____

5. Simple rules everywhere.

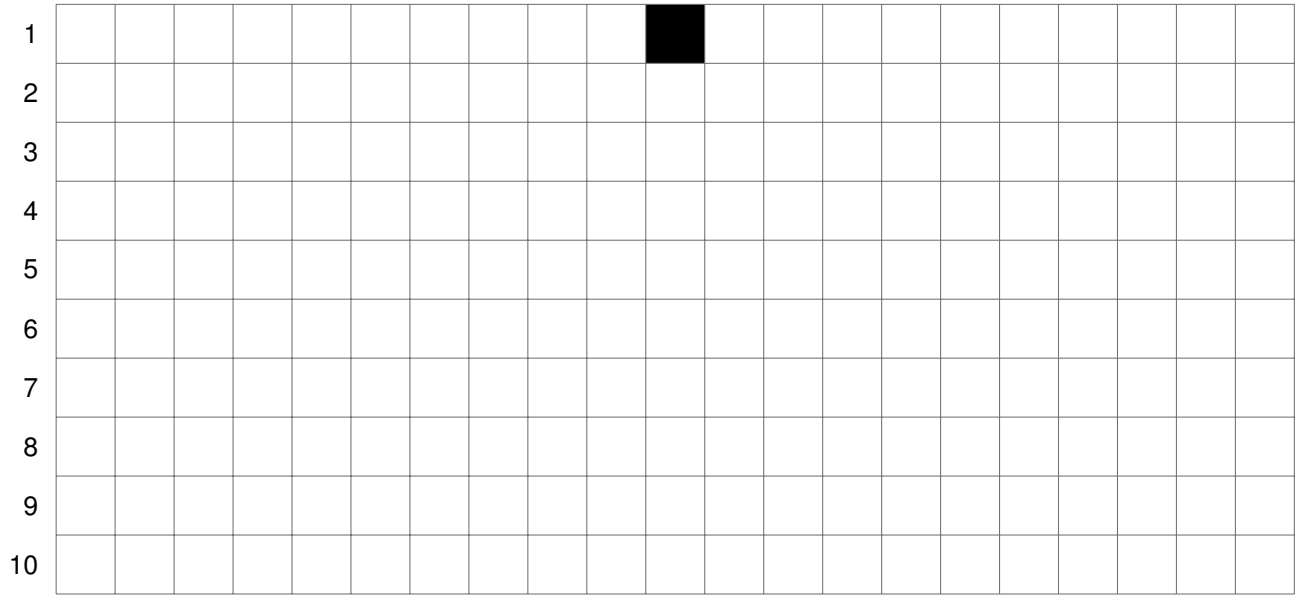
Without "fly the same way," the birds _____

In the Game of Life, did any single cell move? _____ So what moved? _____

6. Talk about it. Can simple parts make something smart?

I think simple rules **can** / **cannot** make something smart, because _____

Go Further (Day 2). Be Rule 30 as a class. Color each new row the class shows.



Game of Life. A black cell with _____ black neighbors stays black. A white cell with exactly _____ black neighbors turns black.